Heart Rate Estimation from Multi-site Wearable PPG

CNN-BiLSTM: data pipeline, model architecture, training, selection and reliability

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- Data statistics

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- Final results (out-of-fold CV)

Task, data and split

Task

one BPM value per 10 s window (regression)

Data

Multisite-PPG + WF-PPG; exercise excluded (only 38 % of windows had usable ECG labels)

Split

by subject: stratified 5-fold CV with rotating validation group; 5 fold cross validation

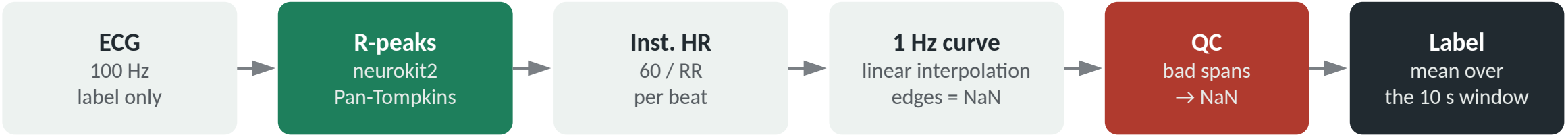
Data statistics

	Multisite-PPG	Wrist-finger-PPG	All
Subjects	20	27	47
Recordings	519 (earring, necklace, ring, watch)	165 (6 trials per subject)	684
Setting	free-living	short controlled clips	–
Recording length	median 4.5 h; 10 % > 23 h	median 4.0 min (1.8-7.0 min)	–
Windows (10 s, 2 s stride)	37 668 (cap 2000 per subject)	14 196	51 864
Windows per subject	median 2000 (474-2000)	median 541 (118-675)	–
Label BPM, mean ± SD	83.3 ± 14.6	79.2 ± 10.4	82.2 ± 13.7
Label BPM, 5th-95th pct.	59.7-107.0	64.1-97.9	61.0-104.5
Label BPM, range	42.5-179.5	56.4-113.3	42.5-179.5
Share < 60 / 60-90 / ≥ 110 BPM	5.3 / 66.8 / 3.9 %	0.7 / 83.9 / 0.1 %	4.0 / 71.5 / 2.8 %

Windows: after ECG quality control, as cached (the model's data). Lengths from the EDF headers. The exercise set (19 files, 8 subjects) is excluded.

Takeaway: multisite gives 73 % of the windows and almost every heart rate above 110 BPM; wrist-finger is short, controlled and sits at 60-90 BPM.

Ground truth: ECG R-peaks to a 1 Hz heart-rate curve



RR checks (per beat)

- Instantaneous HR outside 30-220 BPM
- Neighbouring RR intervals differ by more than × 1.2
- The span between the two peaks becomes NaN

Step / hold interpolation rejected: it biases the window mean toward the first beat.

ECG quality gate

- bSQI: Pan-Tompkins vs neurokit beats matched within 150 ms; 10 s block < 0.8 → NaN
- Flat ECG: cleaned peak-to-peak ≤ 0.1 × recording median → NaN

Catches fake ~180 BPM beat trains on lead-off ramps that pass the RR checks.

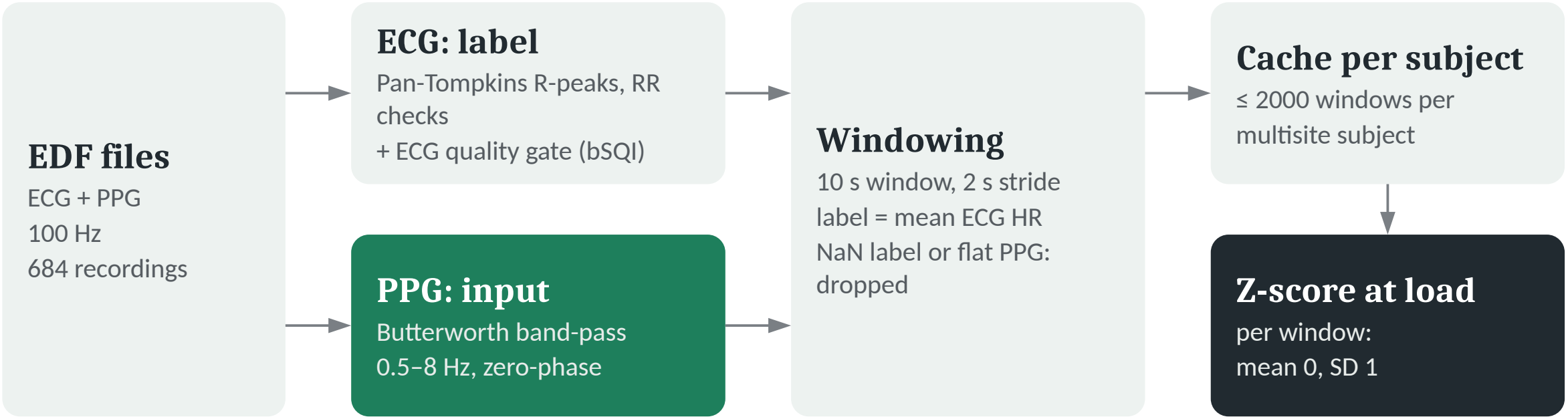
Window rule

- Any NaN second in the window → window dropped
- Raw PPG constant (sensor off, padding) → dropped
- One rule for all QC paths; no interpolation across a bad beat

Wrist-finger keeps 99.7 % of valid seconds.

Finding: most remaining multisite disagreement is motion-corrupted PPG under a correct ECG label, not a wrong label.

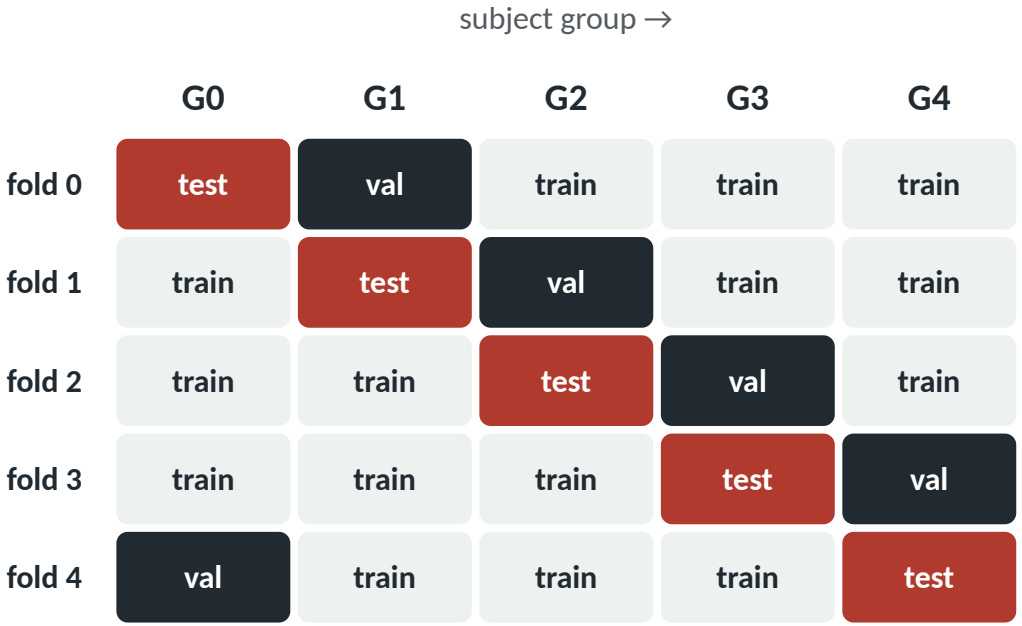
Data pipeline: from EDF to training windows



Label: the ECG says what the heart did; the network only ever sees the PPG.

Cap: multisite recordings last from minutes to several days, so without a cap a few subjects would dominate training.

Subject-wise 5-fold cross-validation



Stratified groups (split.assign_groups)

- Per sub-dataset, score = share of a subject's windows ≥ 130 BPM (multisite) or ≥ 100 BPM (wrist-finger)
- Sort by score, cut into strata of 5, shuffle within a stratum (seed 42), deal to groups 0-4
- Multisite 4 subjects per group, wrist-finger 6 / 6 / 5 / 5 / 5

Rotating validation (split.fold_split)

- Fold k: test = group k, val = group (k+1) mod 5, train = the other 3
- Every subject is in test once and in validation once

Computed once

- make_folds.py → cache/folds.json; sha256 = folds_hash in every run sidecar
- Bootstrap quotas and smoother q use fold train only

Why: the fixed split had only 3 multisite subjects in test, and 6 MSE seeds spread wider (6.08 ± 0.79 BPM) than the whole config table.

Augmentation and balancing

Gaussian noise

σ = 0.05 (z-scored units) added to every training window, freshly drawn each time it is sampled.

Validation and test windows are never augmented.

Balanced sampling

Train set: 28 000 multisite vs 9 918 WF-PPG windows.

WeightedRandomSampler, weight 1 / n per sub-dataset → each sub-dataset fills about half of every epoch.

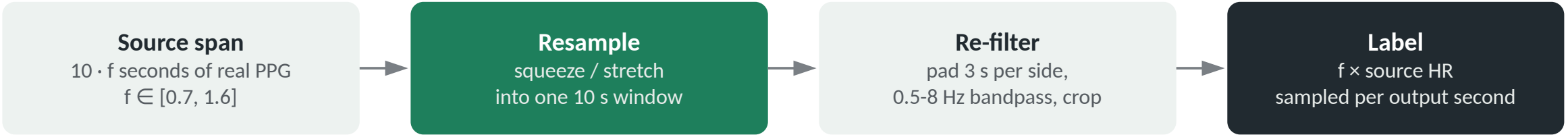
Per-subject cap

At most 2000 windows per multisite subject (random, seed 42).

Keeps long free-living recordings from dominating.

Kept light on purpose: cutout and amplitude scaling stay deferred. Time warping now exists as the opt-in bootstrap (next slide).

Time-warp bootstrap: synthetic low and high heart rates



Set	BPM bins	Sub-datasets
L	40-50, 50-60	both
H	110-190 (10-BPM bins)	multisite
H, --fold	+ 110-130	wrist-finger too

In the sampler (dataset.sampler_weights)

- Synthetic windows get at most 20 % of their sub-dataset's weight
- Epoch length stays the real-window count, so runs compare fairly

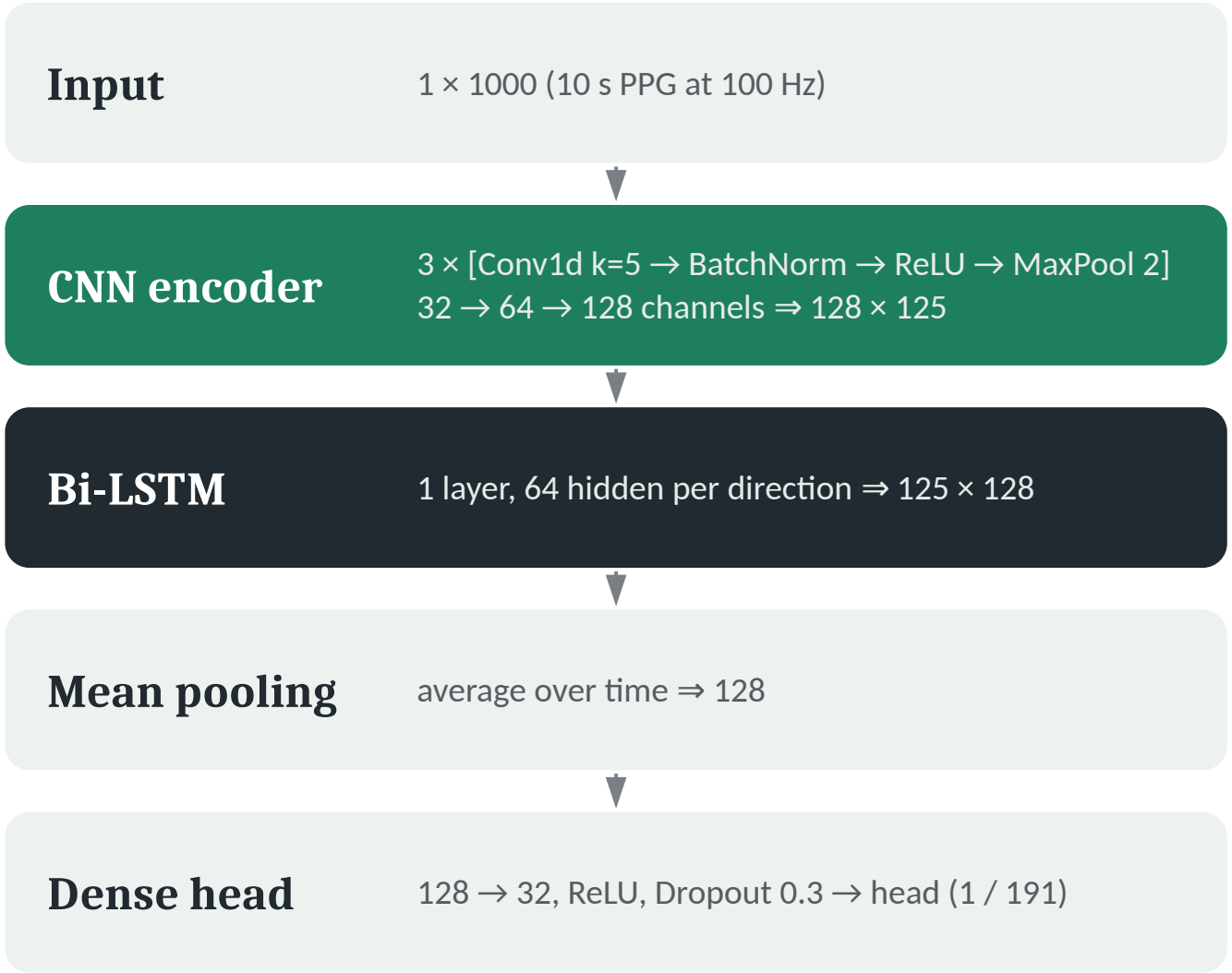
Amount and leakage

- Each bin topped up to 500 windows per sub-dataset, real windows counted
- Each source start reused at most 5 times; short bins flagged, never padded
- Fold train subjects only

Source filter and caveat

- H: only sources whose PPG peak is within 10 % of the label
- Re-filter: no out-of-band energy gives the warp away
- Caveat: warp scales systole too; real high HR shortens diastole

Model: CNN-BiLSTM backbone



CNN extracts pulse morphology (upstroke, notch) and suppresses noise; pooling shortens the sequence 8×.

Bi-LSTM reads the beat sequence forwards and backwards, so a corrupted beat can be inferred from its neighbours.

Mean pooling + dense head turn the whole window into one heart rate value.

CNN-BiLSTM layer by layer

#	Block	Operation	Output	Params
0	Input	10 s bandpass PPG at 100 Hz, z-scored	1 × 1000	–
1	Conv block 1	Conv1d 1→32, k = 5, same · BN · ReLU · MaxPool 2	32 × 500	256
2	Conv block 2	Conv1d 32→64, k = 5, same · BN · ReLU · MaxPool 2	64 × 250	10 432
3	Conv block 3	Conv1d 64→128, k = 5, same · BN · ReLU · MaxPool 2	128 × 125	41 344
4	Bi-LSTM	1 layer, 64 hidden per direction, batch-first	125 × 128	99 328
5	Mean pool	average over the 125 time steps	128	–
6	Dense	Linear 128→32 · ReLU	32	4 128
7	Dropout	p = 0.3 (train only; MC dropout at eval)	32	–
8	Head	Linear 32→1 reg / 32→191 dist	1 / 191	33 / 6 303
	Total	dist 161 791 (only the head differs)		**155 521 (reg)**

Receptive field

- Same padding: only the 3 pools shorten time, 1000 → 125
- One LSTM step covers 36 input samples (0.36 s), steps 80 ms apart
- That resolves upstroke, peak and dicrotic notch of a pulse
- 64 % of parameters in the Bi-LSTM, 33 % CNN, 3 % head
- Mean pool matches the label: mean HR over the window

Output heads: one backbone, two ways to read it

reg (default)

Layer: Linear 32 → 1

Output: BPM directly

Loss: Huber δ, MAE or MSE

Uncertainty: none from one model; seed-ensemble spread, MC-dropout std

In the runs: Stage A (which loss) and Stage B (+ bootstrap)

dist (classification)

Layer: Linear 32 → 191, 1-BPM bins 30-220

Target: Gaussian over bin centres, σ = 2 BPM, sums to 1

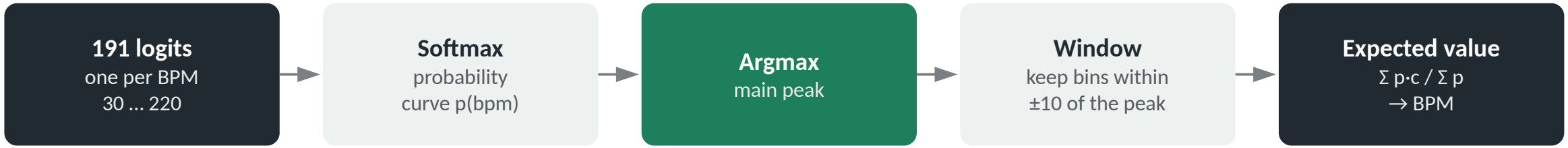
Loss: cross-entropy with probability targets

Output: expected BPM within ±10 bins of the argmax

Uncertainty: std of the softmax curve

In the runs: Stage C (+ bootstrap): the winner

dist head: from logits to one BPM value



Why the ±10 window

- A plain mean over all bins is pulled toward a second mode
- A harmonic (2 × HR) or a motion peak would bias it
- Argmax alone is quantised to 1 BPM; the window keeps it smooth
- Plain argmax is reported too

Training target

- Label y → Gaussian over bin centres, σ = 2 BPM
- Soft target: neighbouring bins share credit
- Cross-entropy per window, then batch mean
- Decoding the target gives back y exactly

Dropped: τ = 1 adjustment

- Prior = smoothed train counts + 1
- The +1 floor lifts never-seen bins (150-220) ~7 logits
- Fold-0 smoke: multisite val MAE 6.4 → 35.6
- τ = 0 only; rare-BPM bias left to the bootstrap

Seed ensemble and uncertainty scores

Combine 3 seeds

reg: mean BPM; spread = std across models

dist: average the softmax curves, decode once

evaluate.py --fold k --ckpt a.pt b.pt c.pt

MC dropout

- Only nn.Dropout on, BatchNorm stays in eval mode
- N = 20 passes (--mc 20)
- Score = std of the 20 predictions
- Evaluation only: no training cost

Prediction file

- subject, dataset_source, split, truth, pred
- scores (primary), score_spread, score_std, score_mc
- NaN where a score does not apply

Score (higher = worse)	Source	Available for
Ensemble spread	disagreement between seeds	every head, ≥ 2 checkpoints
Curve std	width of the dist softmax	dist
MC-dropout std	spread over 20 dropout passes	every head

Quality gate: say "too uncertain" instead of a wrong value

Signal scores (no model)

- −skewness: clean PPG is positively skewed (Elgendi 2016)
- 1 − spectral prominence: power within ±0.1 Hz of the 0.6-3.5 Hz peak

Model scores

- Ensemble spread
- dist curve std
- MC-dropout std

Gate

- ok = score ≤ limit
- Not ok → "value too uncertain"; raw BPM kept
- Limit fitted on the fold's val, applied unchanged to its test

Two operating points

Coverage: limit = 90 % quantile of val scores

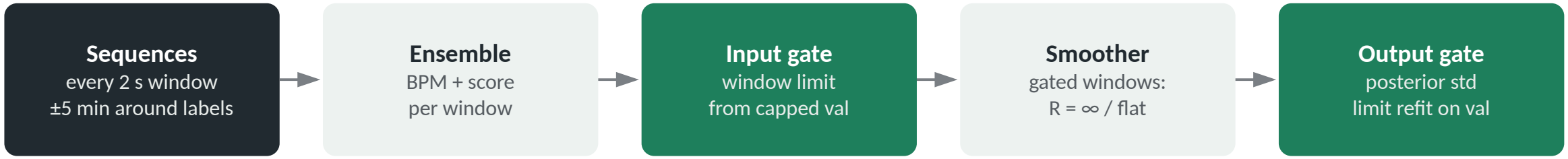
MAE: largest limit with passed val balanced MAE ≤ 5 BPM; infeasible → coverage limit

Subjects without a passed window leave the MAE and are counted.

Checks and report

- Spearman ρ(score, |error|) on val > 0, subject-bootstrap 95 % CI excludes 0
- Risk-coverage curve and AURC
- MAE at 100 / 90 / 80 / 70 % coverage, per bin and sub-dataset

Offline smoothing: pipeline over consecutive windows



Sequences

- predict_sequences.py: all 2 s-stride windows near a labelled one
- 6.6 % of multisite windows are in scope
- Segments split at gaps > 30 s
- Strided view: multisite files reach 9.3 days

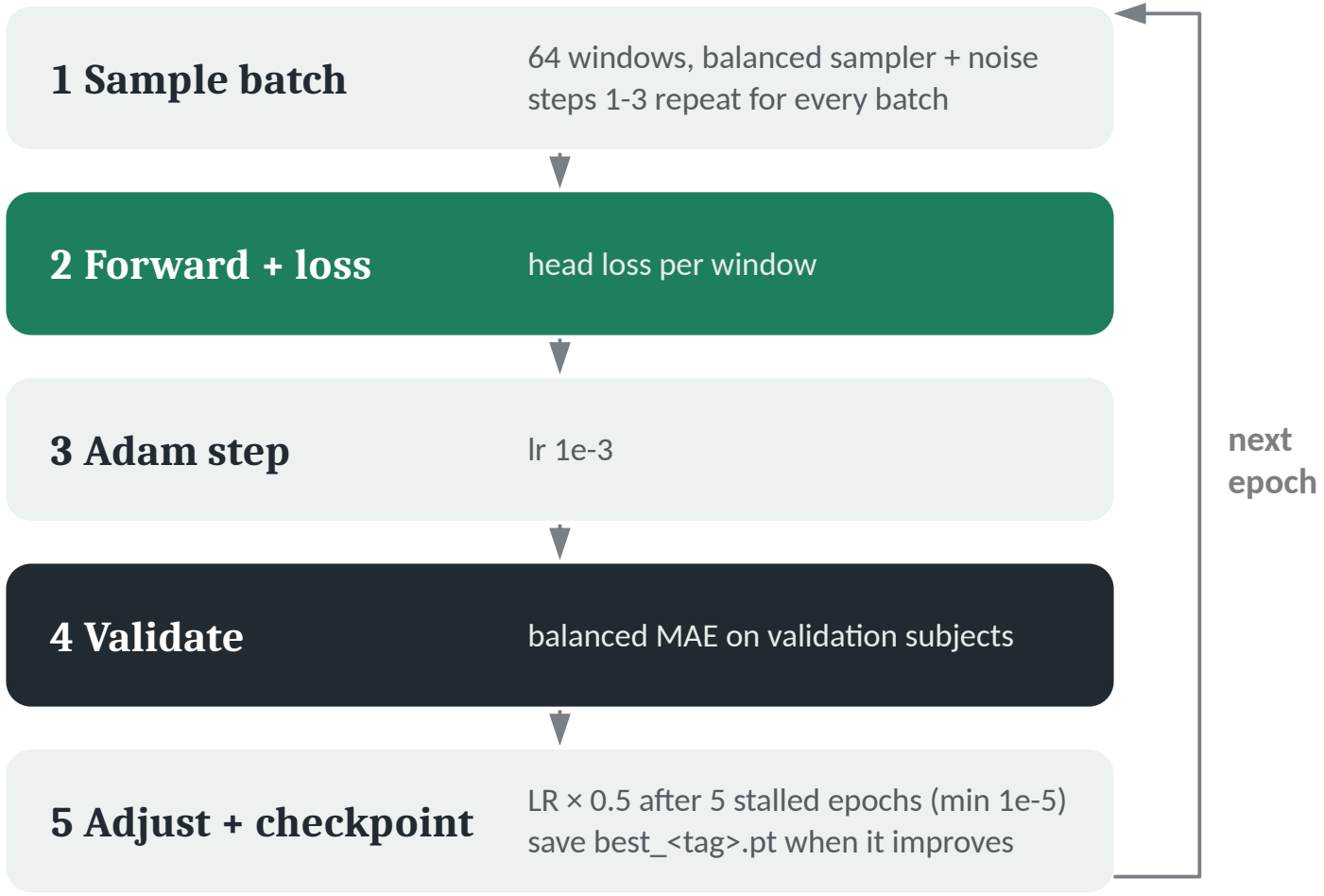
Tuning and scoring

- Tuned on the fold's val group: k, m, c
- Scored on test: all labelled windows, uncapped
- Raw recomputed on the same windows
- Gated runs > 30 s stay uncovered

Report

- Raw vs each smoother, with / without gates
- Subject-macro and pooled MAE, per bin, coverage
- Lag: error in the 30 s after the largest HR jumps

Training loop



Loss: the winner (dist head) uses cross-entropy on Gaussian soft targets; reg runs use MSE, MAE or Huber.

Early stopping: stop after 20 epochs without validation improvement, at most 100 epochs.

Model selection: the checkpoint with the lowest validation MAE is evaluated, not the last epoch.

The question, in three acts

1 Choose the model

- 5 training setups x 5 folds x 3 seeds = 75 runs
- Stage A: loss · Stage B: bootstrap · Stage C: distribution head
- Decided on validation only, checked per test fold

2 The problem it reveals

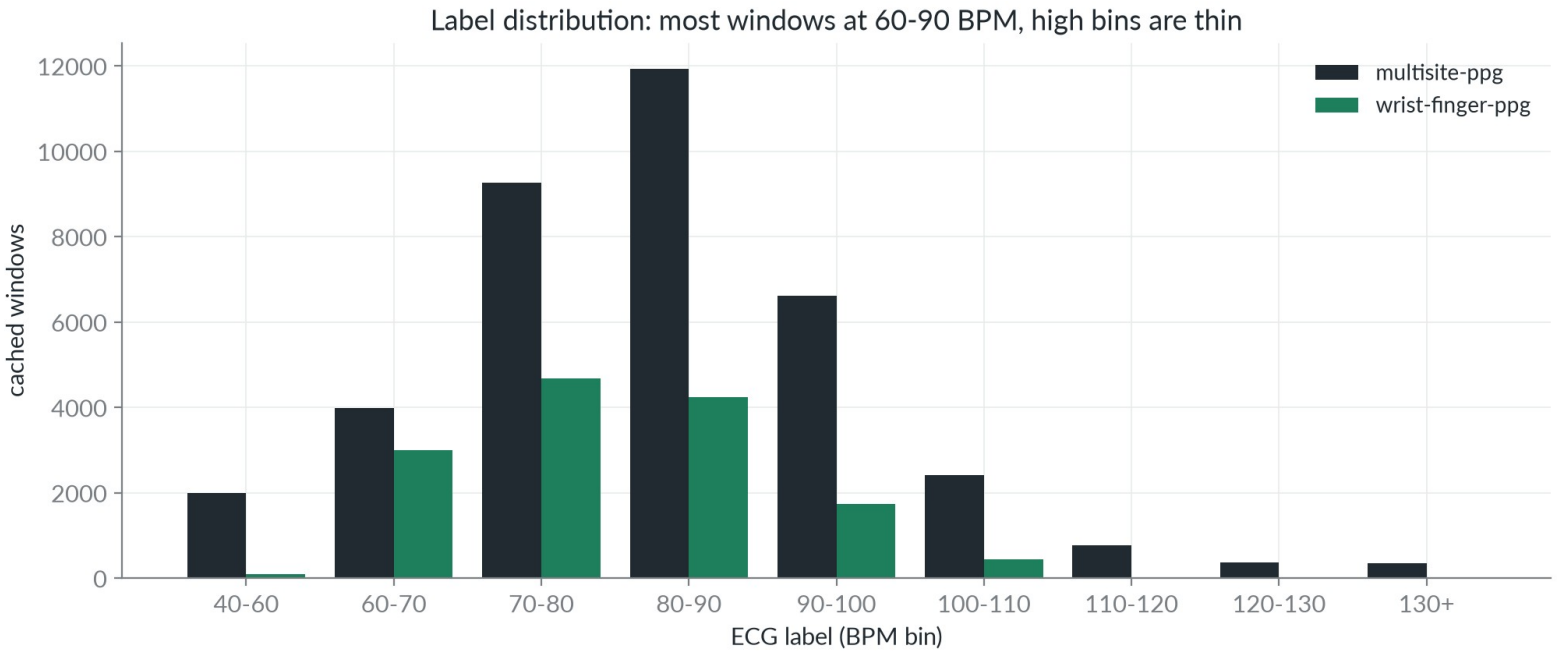
- Every model pulls predictions toward the middle (75-85 BPM)
- High heart rates are under-predicted by tens of BPM
- Root cause: the label distribution

3 Make the output usable

- Stage D: average 3 seeds
- Stage E: drop windows the model is unsure about
- Stage F: smooth along time (Kalman, HMM)

Takeaway: Input: one 10 s PPG window (100 Hz, band-pass 0.5-8 Hz). Output: heart rate in BPM. Truth: ECG R-peaks.

Two datasets, one crowded middle



- **47 subjects**: 20 multisite-PPG (free-living, 4 wearable sites), 27 wrist-finger-PPG (short controlled clips)
- **51,864 windows** of 10 s, 2 s stride, at most 2000 per subject
- **72 %** of all windows have a label of 60-90 BPM
- **2.8 %** are at 110 BPM or more, almost all from multisite

Takeaway: A model that minimises average error can mostly ignore heart rates above 110 BPM. It does, as we will see.

Run plan: one question per stage (all runs done)

Stage	Question	Configs	Runs	Winner
A	Which loss?	Huber δ10, MAE, MSE · reg · no bootstrap	45	MSE
–	The problem	per-bin bias and slope from Stage A predictions	0	pull to middle
B	Fix with data?	L* + time-warp bootstrap HL	15	MSE + boot
C	Fix with the output?	dist head + bootstrap HL	15	dist + boot
D	Ensemble	3-seed ensemble of the B and C winners	0	3 seeds
E	Quality gate	all scores incl. MC dropout (20 passes)	0	curve std
F	Smoothing	median / Kalman / HMM on 2 s sequences	0	Kalman + gates

Every config

- 5 folds × 3 seeds = 15 runs

Final headline

- Single model → + ensemble → + input gate → + smoother → + output gate
- Each layer reported separately

Losses: what moves the weights

Loss	Per window ($e = \hat{y} - y$)	Gradient for large $\lvert e \rvert$	Role
MSE	e^2	grows with e	baseline for all tests (aggregate.BASELINE)
MAE	$\lvert e \rvert$	constant ±1	robust to label errors, but no fine steps near 0
Huber δ	$\frac{1}{2}e^2$ if $\lvert e \rvert \le \delta$, else $\delta(\lvert e \rvert - \frac{1}{2}\delta)$	capped at ±δ	δ = 10 BPM in Stage A (CLI default 5)
Cross-entropy	$-\Sigma\, q(c) \log p(c)$, q = Gaussian target	on the curve, not on e	dist head

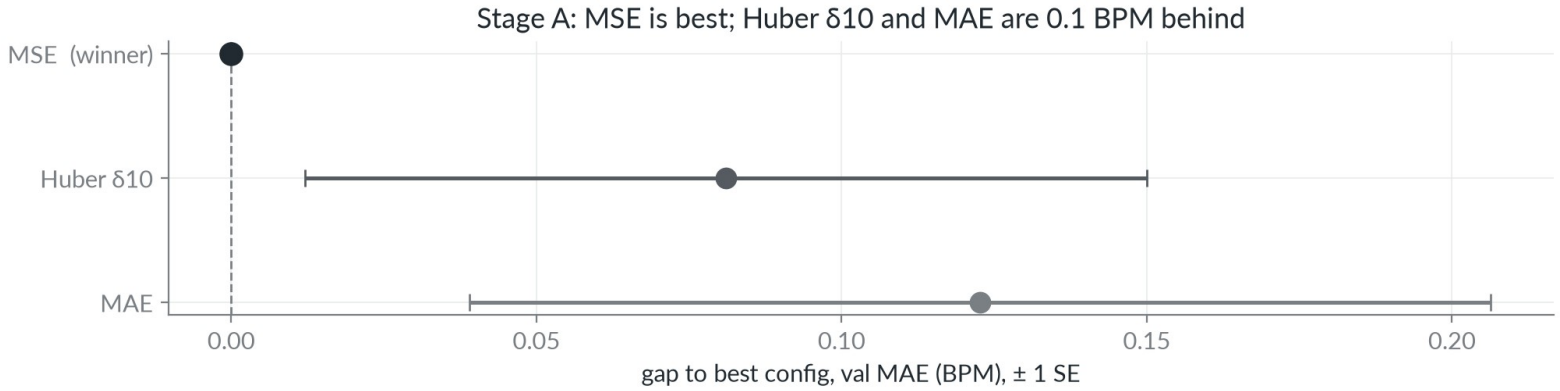
Same rules for every loss

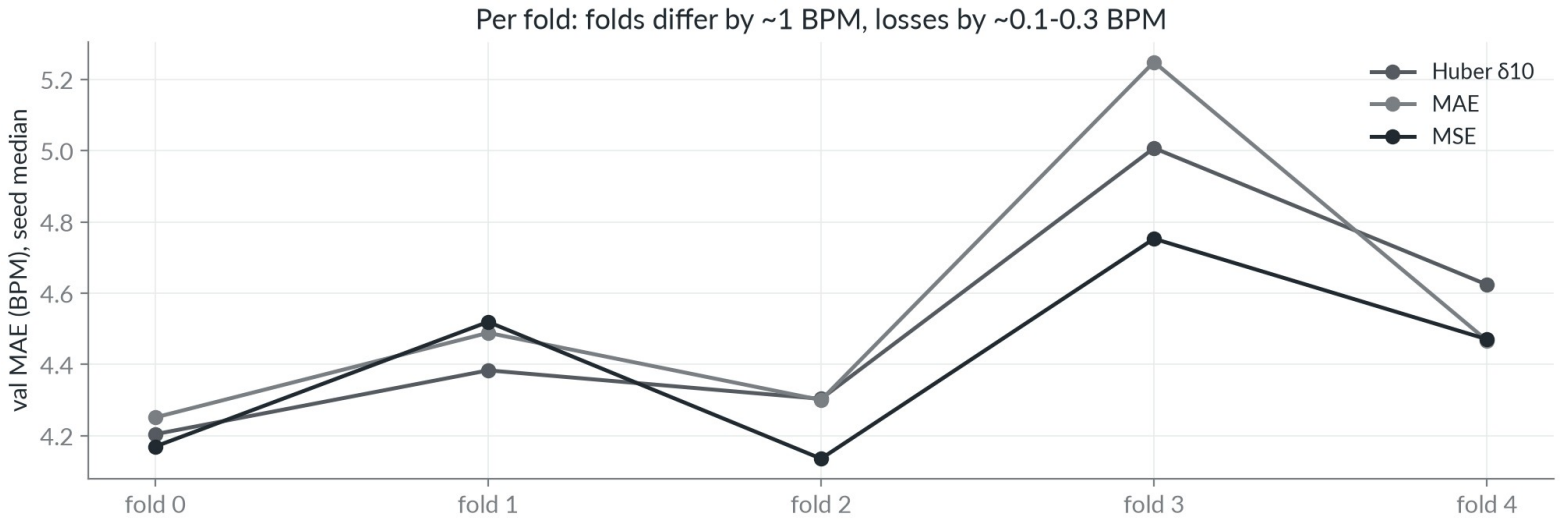
- Per-window loss, then batch mean
- Early stopping and LR schedule on balanced val MAE, never on the loss
- So runs with different losses compare fairly

The question behind Stage A

- Huber and MAE cap how hard one large error pulls
- Good against wrong labels, but it may also ignore real high-HR errors
- Stage A result: MSE wins by ~0.1 BPM; the winner uses the dist head (cross-entropy)

Stage A: MSE wins, by a small margin

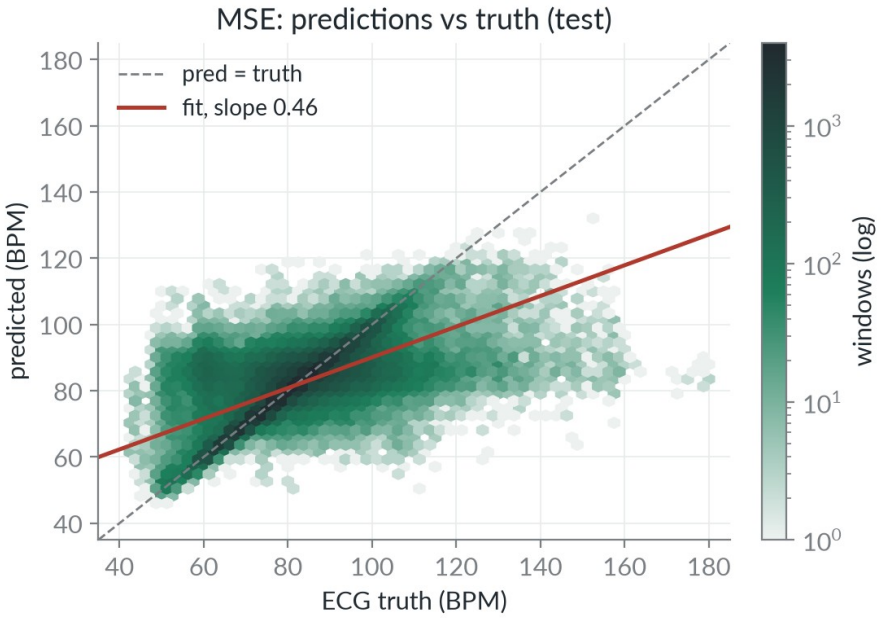


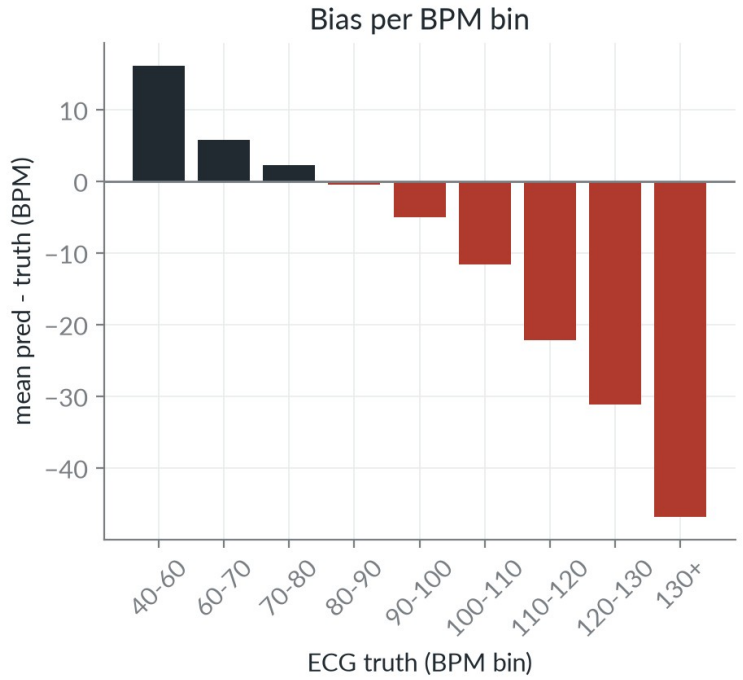


- **MSE wins** globally and in all 5 LTGO checks
- Huber δ10 +0.08 BPM (SE 0.07), MAE +0.12 (SE 0.08)
- Huber δ10 is inside the tie set in 3 of 5 LTGO checks
- **Folds matter more than losses:** per-fold val MAE spans ~4.1-5.3 BPM, losses differ by 0.1-0.3

Takeaway: L* = MSE. The choice of loss is a minor lever.

All models pull toward the middle

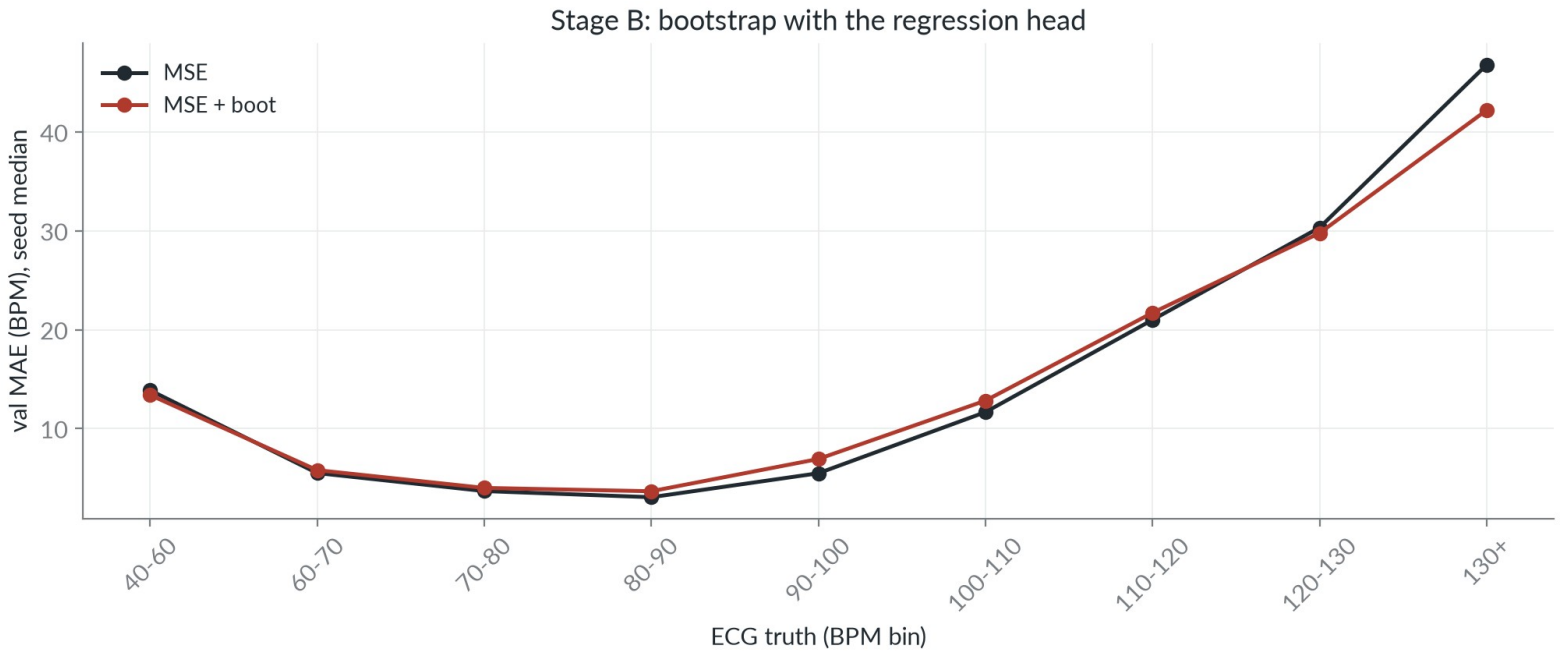




- **Slope 0.46:** a 40 BPM rise in true heart rate moves the prediction by only 19 BPM
- **≥ 120 BPM:** under-predicted by 39 BPM on average (MAE 38.7, n = 2,121 windows)
- **40-60 BPM:** over-predicted by 16.2 BPM
- 70-80 BPM: MAE 3.8, where most data lives
- MSE, test windows of all 5 folds and 3 seeds

Takeaway: Average MAE hides it: the model is good where data is dense and close to useless at high heart rates.

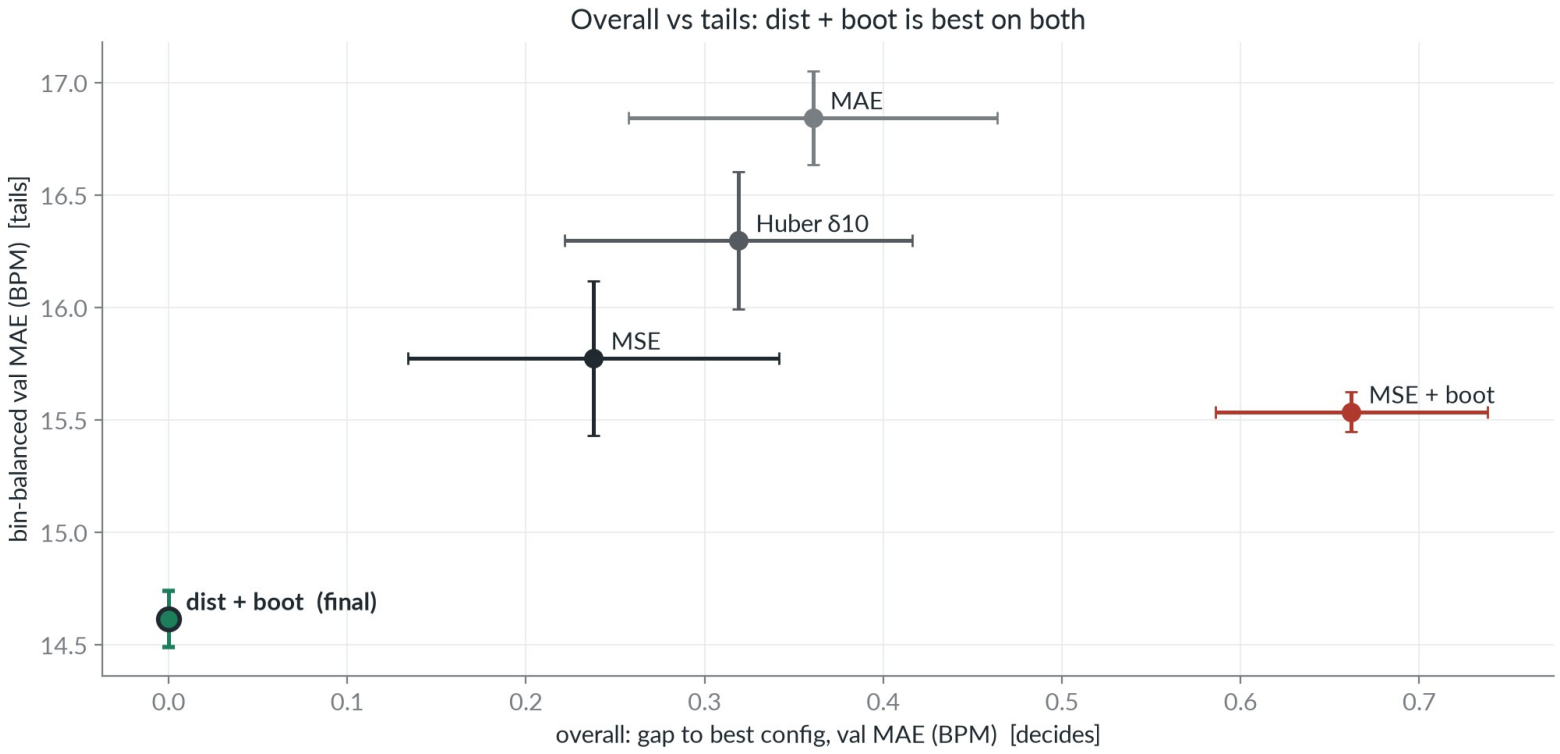
Stage B: the bootstrap alone does not win



- **Bootstrap HL:** time-warped extra windows in the low and high bins
- 130+ BPM error 47 → 42, but 80-100 BPM gets worse (3.1 → 3.7, 5.5 → 6.9)
- Still 0.66 BPM behind plain MSE overall

Takeaway: With a regression head, every BPM gained in the tails is paid for in the middle.

Stage C: the distribution head wins



- **dist head:** 191 bins (30-220 BPM), Gaussian soft targets, BPM = expected value
- **dist + boot is best overall:** MSE is +0.24 BPM (SE 0.10), MSE + boot +0.66 (SE 0.08)
- …and better in the tails than MSE: bin-balanced 14.6 vs 15.8 BPM
- **Final setup: dist + boot**, chosen in all 5 LTGO checks

Takeaway: Lower-left is better. Only dist + boot improves both axes over MSE.

Winner model: dist + boot, 3 seeds, Kalman + gates

Per 10 s window: model

PPG window
10 s at 100 Hz
band-pass 0.5-8 Hz
z-scored

→

CNN-BiLSTM ×3
seeds 42, 1, 2
same backbone
161 791 params each

→

dist head
Linear 32 → 191
softmax over
30-220 BPM

→

Average curves
mean of 3 softmax
decode: ±10 bins
around argmax

Per recording: post-processing

BPM + score
expected BPM
curve std =
uncertainty

→

Input gate
curve std > val limit
→ R = ∞
(no weight)

→

Kalman + RTS
over 2 s windows
$R = (c \cdot \text{curve std})^2$
forward + backward

→

Output gate
posterior std > limit
→ "too uncertain"
else BPM

Backbone

- 3 × [Conv1d k 5 · BN · ReLU · MaxPool 2], 32 → 64 → 128 ch
- Bi-LSTM, 64 hidden per direction
- Mean pool → Linear 128 → 32 · ReLU · Dropout 0.3

dist head

- 191 bins of 1 BPM, 30-220
- Target: Gaussian over bins, σ = 2 BPM
- Output: expected value within ±10 bins of the peak; curve std as score

Size and cost

- 3 × 161 791 = 485 373 parameters
- Only the last layer differs from the reg model
- Gates + smoother: no extra training, tuned on val

Config: --head dist --bootstrap HL, τ = 0; chosen in all 5 LTGO checks.

Winner model: how it is optimised

Training data

- **Time-warp bootstrap HL:** synthetic windows at 40-60 and 110-190 BPM, ≤ 20 % of a sub-dataset's weight
- **Balanced sampler:** each sub-dataset fills half of every epoch
- **Per-subject cap:** ≤ 2000 windows per multisite subject
- **Noise:** Gaussian σ = 0.05 per training window
- **Labels:** ECG R-peaks, 3 QC layers, bad windows dropped

Objective and optimiser

- **Loss:** cross-entropy against a Gaussian soft target (σ = 2 BPM)
- **Adam:** lr 1e-3, batch 64
- **LR schedule:** × 0.5 after 5 stalled epochs, min 1e-5
- **Early stop:** 20 epochs without val gain, max 100
- **Checkpoint:** best window-pooled balanced val MAE
- **Seeds:** 42, 1, 2 (init, shuffle, noise)

Inference and reliability

- **Ensemble:** mean of the 3 softmax curves, decoded once
- **Quality gate:** limit = 90 % quantile of val curve std
- **Kalman + RTS:** m ∈ {1, 2, 4}, c ∈ {0.5, 1, 2, 4} tuned on val
- **Output gate:** posterior-std limit refit on val
- **Tested, not used:** τ = 1 prior, MC dropout as gate

Everything fitted (bootstrap quotas, gate limits, smoother parameters) uses the fold's train or val subjects only.

Results: winner model, out-of-fold test

dist + boot · 3-seed ensemble · input gate · Kalman + RTS: MAE 3.64 BPM

Subject-macro MAE, mean of the 2 sub-datasets, all 47 subjects, 428k labelled 10 s windows (every 2 s)

Raw ensemble: multisite 6.74 · wrist-finger 1.48 · both 4.11 BPM (RMSE 7.94)

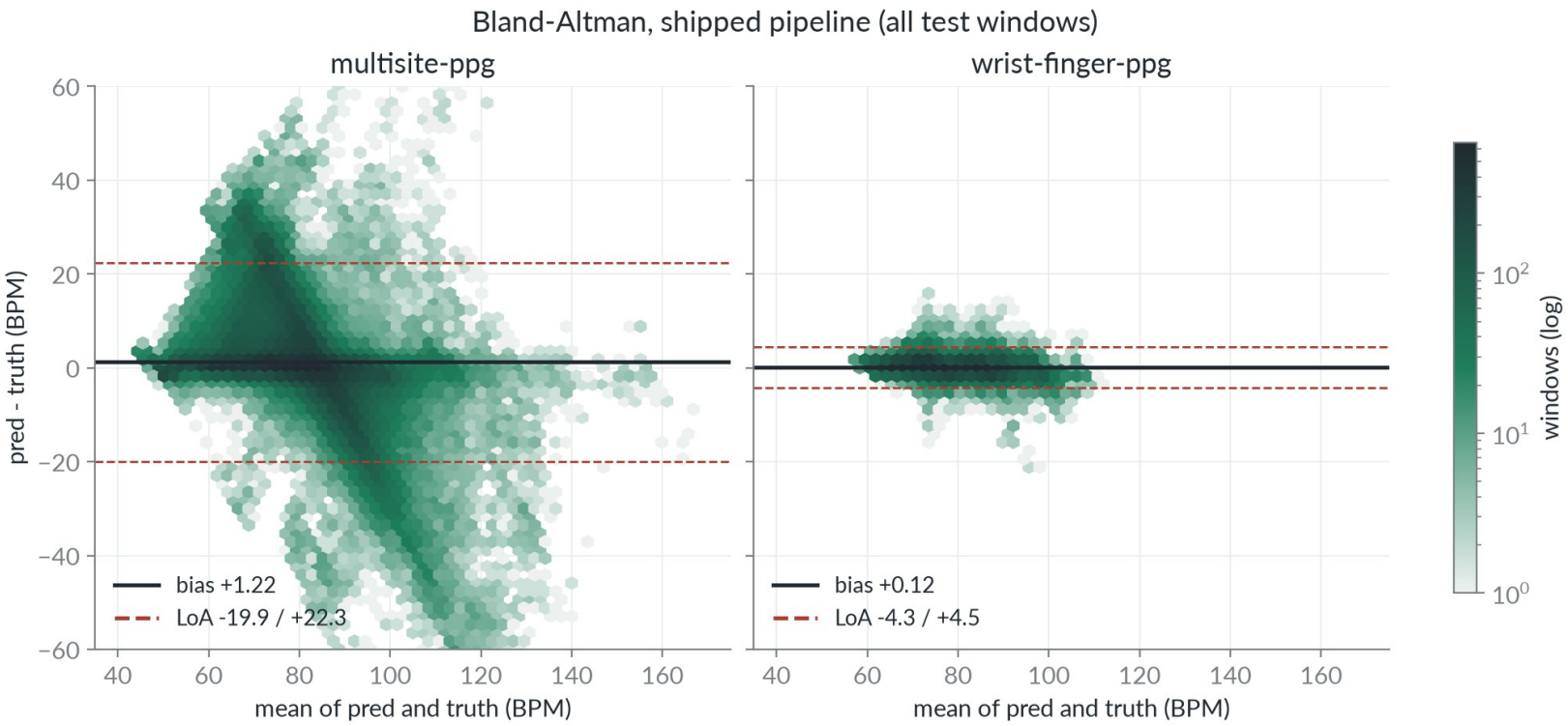
Shipped pipeline, every window: multisite 5.82 · wrist-finger 1.47 · both 3.64 BPM (RMSE 6.54)

Shipped, gated gaps ≤ 30 s only (98.5 %): multisite 5.66 · wrist-finger 1.47 · both 3.56 BPM

Within 5 BPM: multisite 63 % · wrist-finger 96 % of windows; within 10 BPM: 77 % · 99.5 %

Protocol: Out-of-fold: each subject is tested once, by the 3 seeds of the fold whose test group it is in (never trained on it); gate limit and Kalman parameters tuned on that fold's val group. The submitted model has no test data left; these numbers are its estimate.

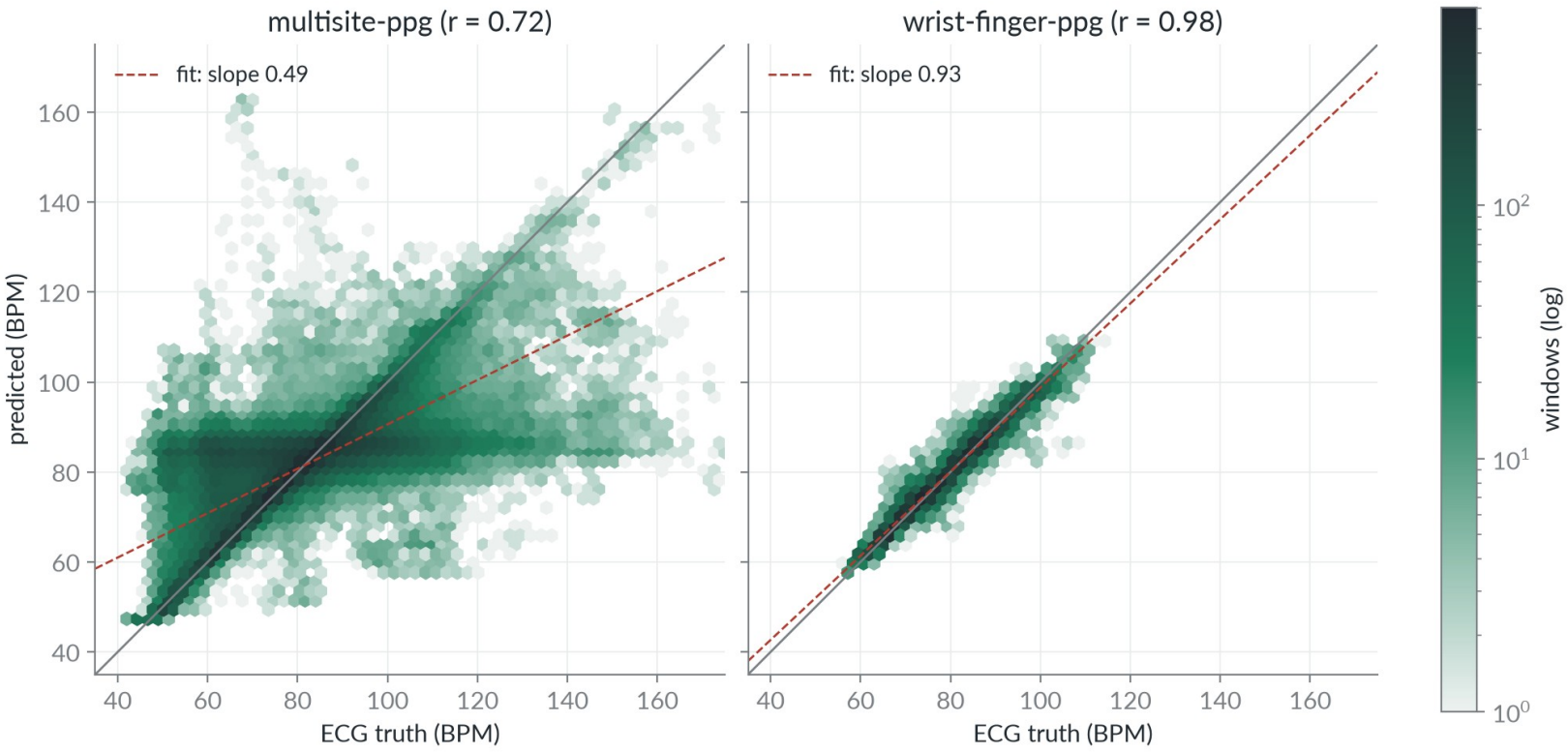
Bland-Altman: unbiased, wide limits on multisite



- **Wrist-finger:** bias +0.12 BPM, 95 % limits −4.3 / +4.5
- **Multisite:** bias +1.2 BPM, limits −19.9 / +22.3
- **Diagonal bands:** prediction stuck near 80 BPM while the true HR moves (the pull to the middle)
- **Data:** all labelled test windows, shipped pipeline, density on log scale

Takeaway: Controlled finger/wrist clips are near ECG quality; free-living multisite errors are large but not systematic on average.

Predicted vs ECG: the slope tells the story



- **Wrist-finger:** r = 0.98, slope 0.93
- **Multisite:** r = 0.72, slope 0.49: a horizontal band at 80-90 BPM
- **Raw vs shipped:** multisite r 0.57 vs 0.72 (gate + Kalman)
- **Grey line:** perfect agreement; red dashed: least-squares fit

Takeaway: Where the PPG has no clear pulse the model falls back to the typical heart rate.

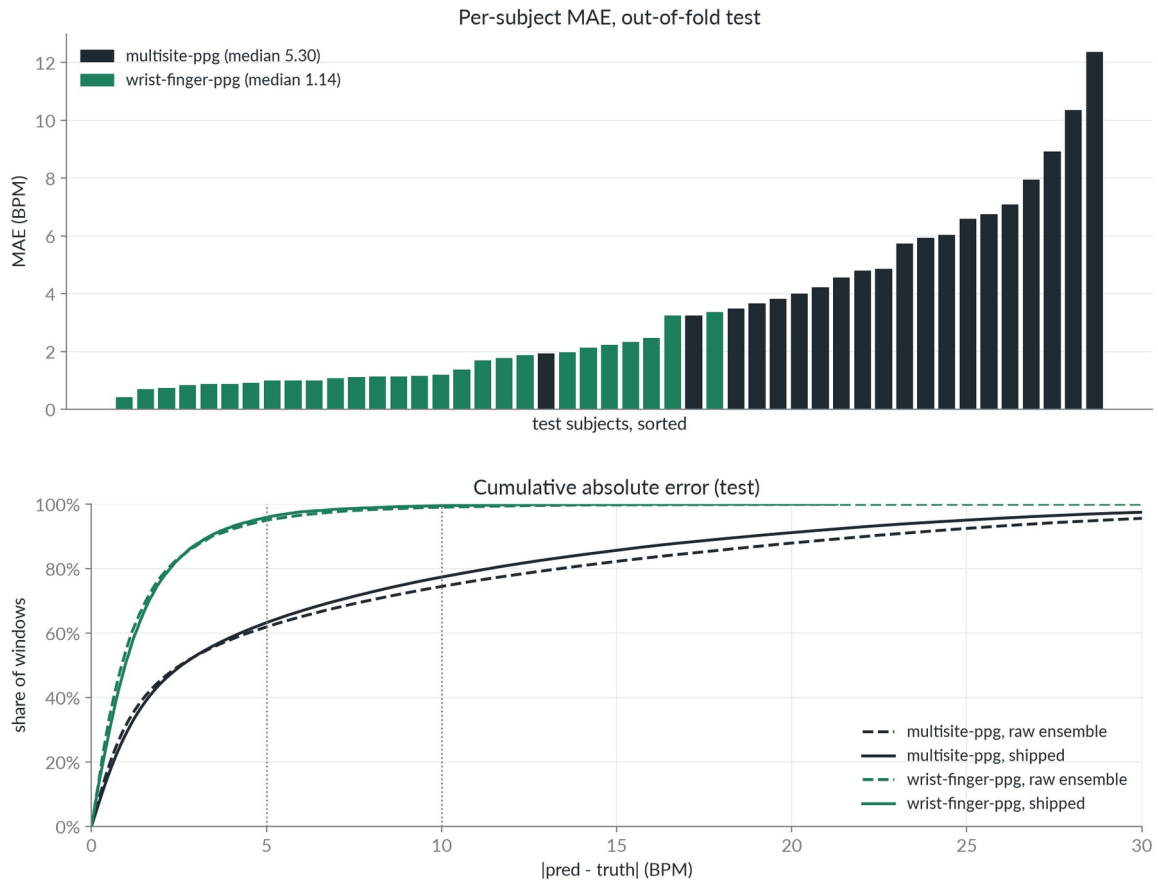
Error per heart-rate bin: the tails stay hard



- **80-90 BPM:** MAE 2.2, bias −0.3
- **40-60 BPM:** MAE 11.8, over-predicted by 11.4
- **130+ BPM:** MAE 42.5, under-predicted by 42.4 (2,820 windows)
- **Smoothing:** lowers MAE in every bin by about 1 BPM, cannot fix the bias

Takeaway: The bootstrap and the dist head narrowed the pull to the middle; they did not remove it.

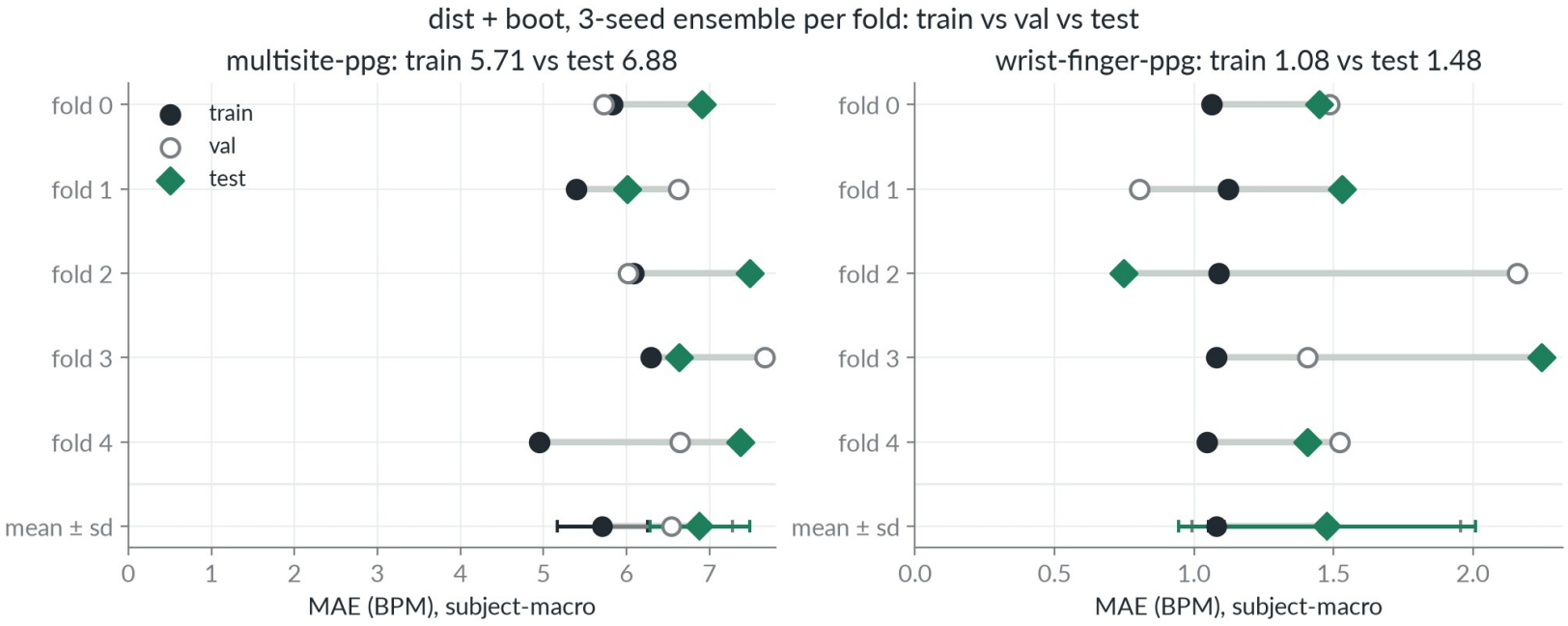
Per subject: wrist-finger uniform, multisite spread



- **Median subject MAE:** wrist-finger 1.14, multisite 5.30 BPM
- **Worst subject:** multisite P14, 12.4 BPM
- **Every wrist-finger subject:** ≤ 3.4 BPM
- **Within 5 BPM:** 96 % (wrist-finger), 63 % (multisite) of windows

Takeaway: Multisite error is spread over many subjects, not one outlier.

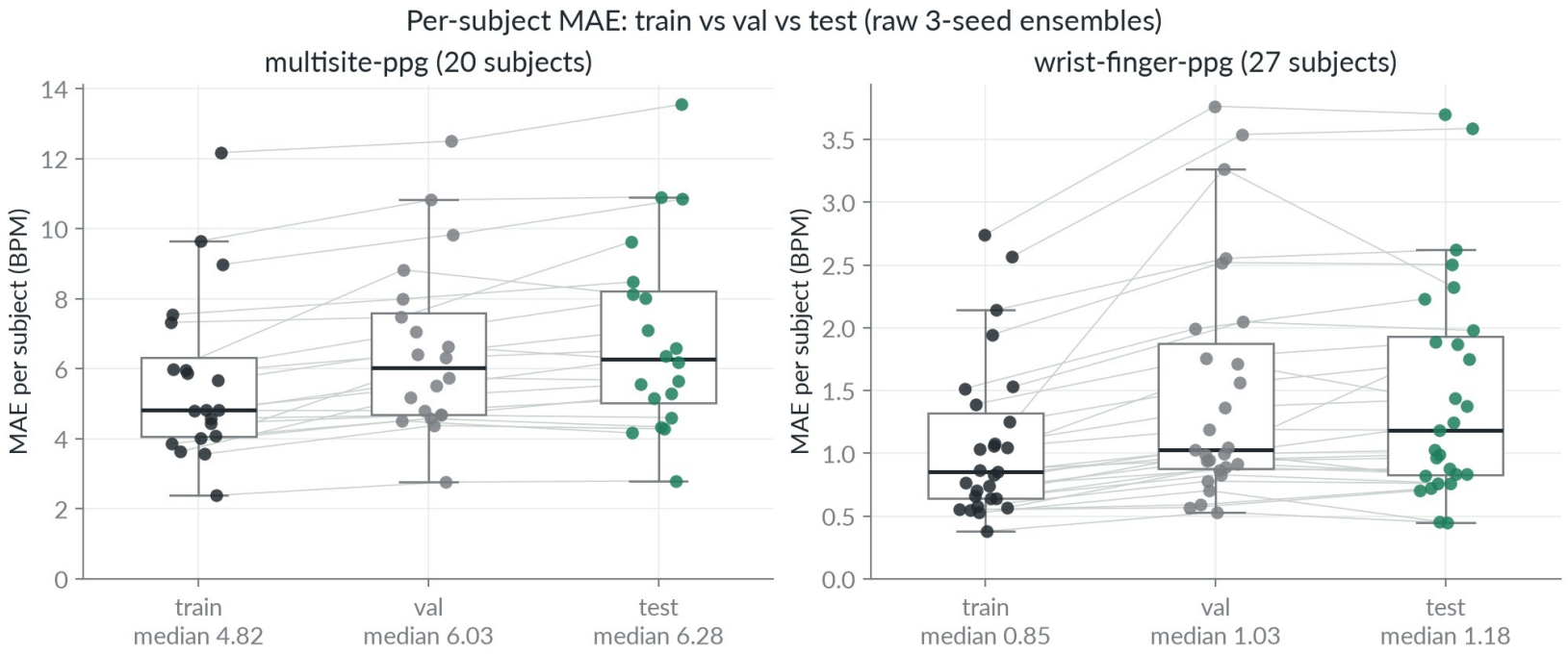
Train vs test: 5 fold ensembles



- **Both datasets:** train 3.39 ± 0.27, val 4.00 ± 0.37, test 4.18 ± 0.27 BPM (mean ± sd over folds)
- **Multisite:** train 5.71 vs test 6.88
- **Wrist-finger:** train 1.08 vs test 1.48
- **Gap:** about 0.8 BPM; even on train, multisite stays near 6 BPM

Takeaway: Mild over-fitting; multisite error is mostly PPG quality, not memorisation.

Train vs test per subject



- **Median subject MAE:** multisite 4.82 / 6.03 / 6.28, wrist-finger 0.85 / 1.03 / 1.18 (train / val / test)
- **Lines:** the same subject across splits
- **Hard subjects** stay hard in every split

Takeaway: The ranking of subjects barely changes between train and test.